

IBM University Engagement Project Submission

“Exploring the Power of Machine Learning with IBM Cloud”

Project Title – Loan Approval Prediction using Machine Learning with IBM watsonx.ai AutoAI

Domain of Project – Banking & Finance

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
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Problem Statement: Loan Approval Prediction using Machine Learning

Financial institutions receive a large number of loan applications every day from individuals with diverse financial and personal backgrounds. Evaluating each application manually is a time-consuming and error-prone process that may lead to inconsistent decisions, delayed approvals, and increased operational costs. Incorrect loan approval decisions can result in financial losses for banks, while rejecting eligible applicants may reduce customer satisfaction. The objective of this project is to design, build, and evaluate a **binary classification model** that accurately predicts whether a loan application should be **approved or rejected** based on applicant information such as gender, marital status, education, employment status, income, loan amount, loan term, credit history, and property area. The goal is to develop a reliable machine learning solution using **IBM watsonx.ai AutoAI** that assists financial institutions in making faster, more accurate, consistent, and data-driven loan approval decisions.

Proposed Solution: (IBM watsonx.ai Studio – AutoAI)

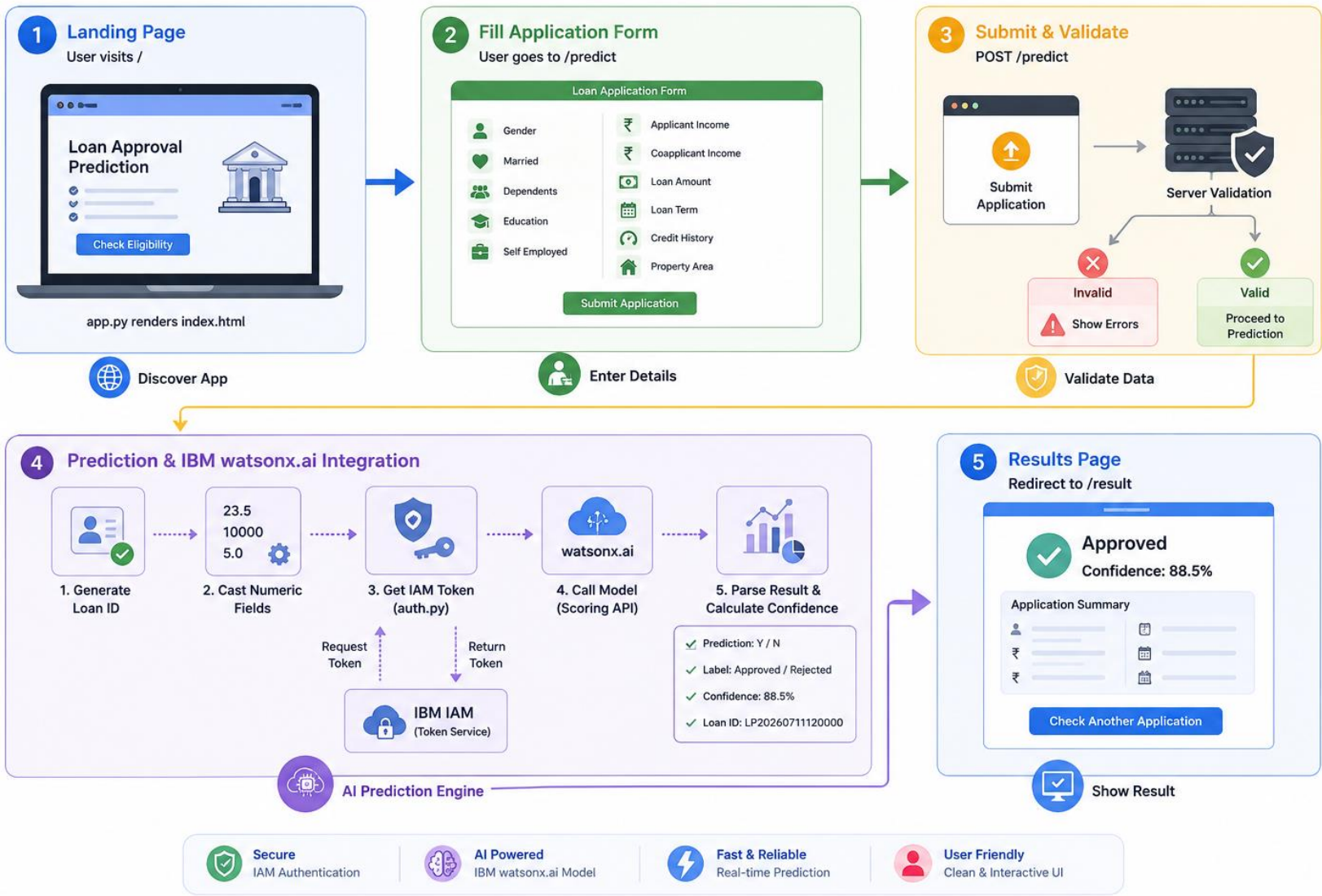
I propose an **AI-powered Loan Approval Prediction System** using **IBM watsonx.ai Studio** and **AutoAI**. Applicant information such as **gender, marital status, education, employment status, applicant income, co-applicant income, loan amount, loan amount term, credit history, and property area** is uploaded to **watsonx.ai**. **AutoAI** automatically performs **data preprocessing, dataset splitting, feature engineering, algorithm selection, hyperparameter optimization, and ensemble creation** to generate and evaluate multiple machine learning pipelines. The generated pipelines are ranked based on **cross-validation accuracy**, and the **best-performing model** is automatically selected for deployment. The selected model predicts whether a loan application will be **Approved (Y)** or **Rejected (N)**, enabling financial institutions to make **faster, more accurate, consistent, and data-driven lending decisions**.

Technology Used

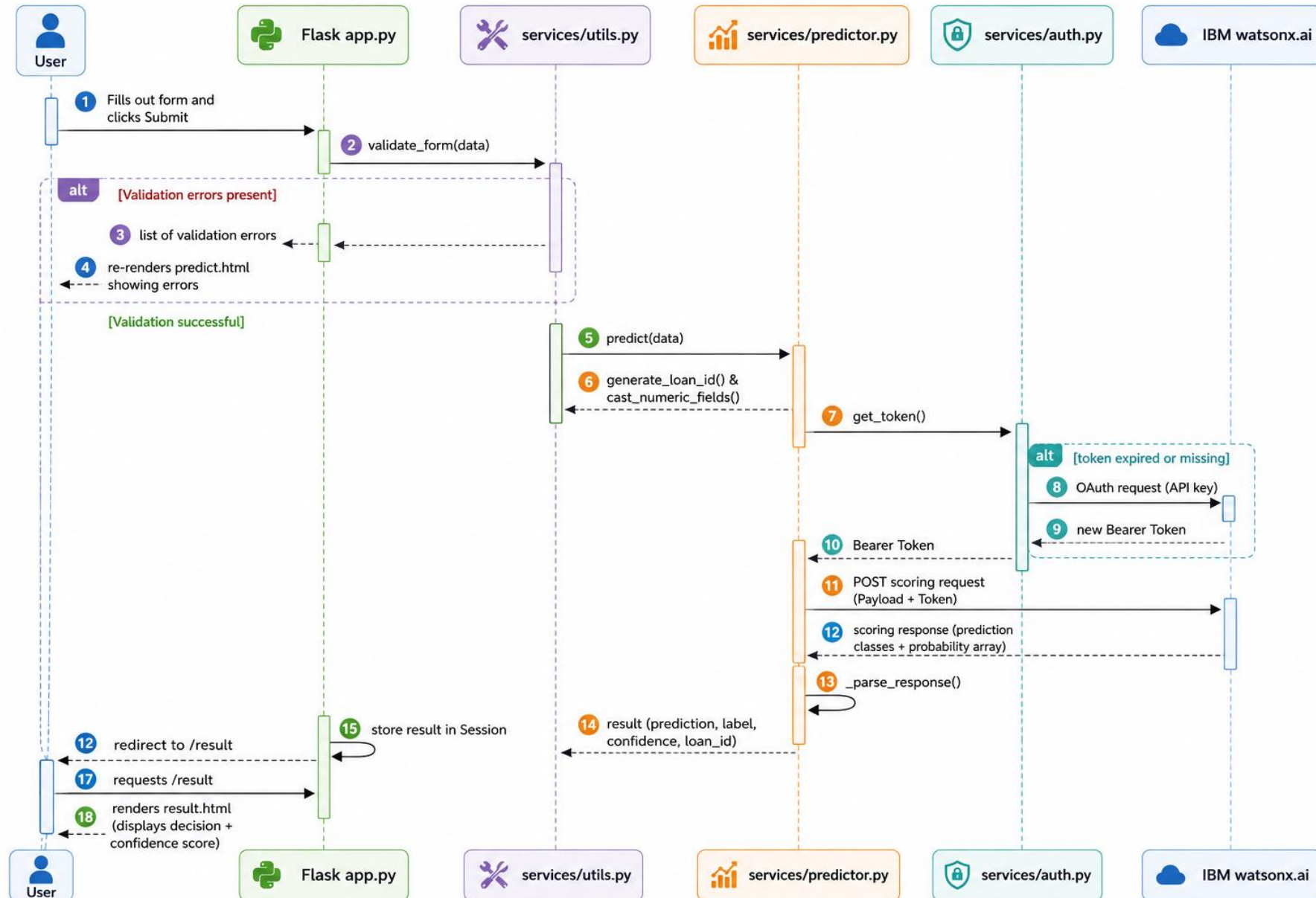
- **IBM Cloud** – Provides a secure, reliable, and scalable cloud platform for developing, training, and deploying machine learning solutions.
- **IBM watsonx.ai Studio (AutoAI)** – Used to automate the machine learning workflow, including data preprocessing, feature engineering, model selection, hyperparameter optimization, and pipeline generation.
- **Programming Language:** Python (Auto-generated notebook from IBM watsonx.ai)
- **Dataset:** Loan Approval Prediction Dataset
- **ML Algorithm / Model Used:** Batched Tree Ensemble Classifier (Snap Random Forest Classifier)
- **Machine Learning Task:** Binary Classification
- **Prediction Column:** Loan_Status
- **Accuracy Obtained:** 81.0% (Cross-validation Accuracy)
- **Enhancements & Hyperparameter Tuning:** Feature Engineering (FE), Hyperparameter Optimization (HPO-1 & HPO-2), Batch Ensemble Creation
- **IBM Bob**

Website Workflow Diagram

LOAN APPROVAL WEB APP – WORKFLOW

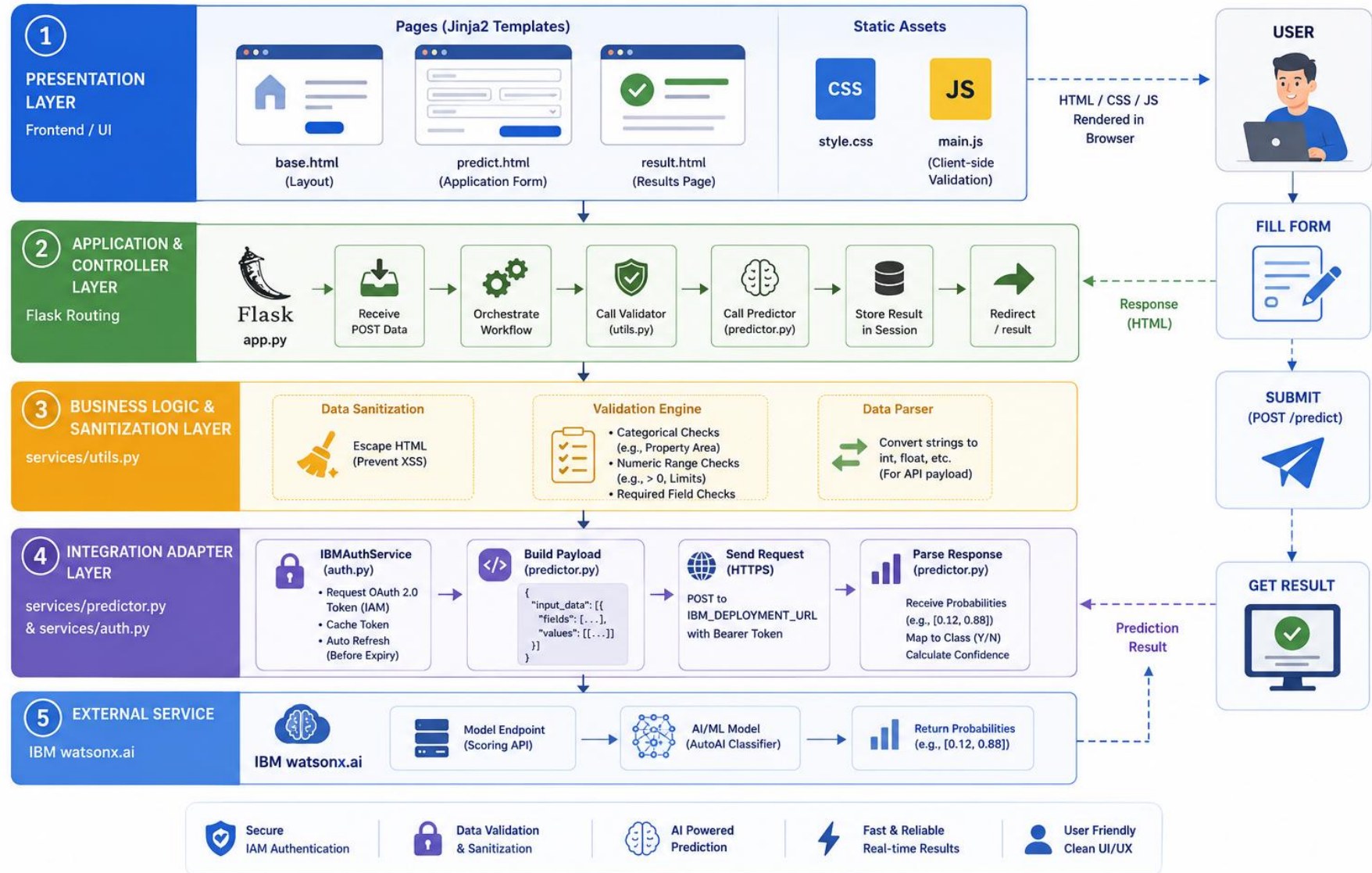


Website Workflow Diagram



Website Architecture Diagram

LOAN APPROVAL WEB APP – ARCHITECTURE WORKFLOW



Training Screenshots

1 Project Creation

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

1

Rahul Karunya Garige's Acc...

Frankfurt

RG

Projects / Loans Application

Launch IDE

Build machine learning models automatically

Define the details to create an AutoAI experiment asset and open it in the AutoAI tool.

+ New

Sample

Define details

Name

Loans Application

Description (optional)

What's the purpose of this AutoAI experiment?

Tags (optional)

Add tags to make assets easier to find.

Start typing to add tags

Define configuration

watsonx.ai Runtime service instance

WatsonMachineLearning

Environment definition

Large: 8 CPU and 32 GB RAM

This environment definition consumes **20 capacity units per hour** for training. For details, see [watsonx.ai Runtime plans](#).

Cancel

Back

Create

Training Screenshots

2 Importing Dataset

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

?

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Projects / Loans Application / Loans Application

Configure AutoAI experiment

Loans Application 

Add data source



Drop data files here or browse for files to upload

Add files such as tabular data (CSV).

Browse

Select from project

Training Screenshots

3 Importing Dataset (Successful)

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

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Projects / Loans Application / Loans Application

Configure AutoAI experiment

Loans Application

Autosaved: 15:44:51

Add data source

Add files such as tabular data (CSV).

Browse

Select from project

5 Loan Approva lPrediction.csv

Size: 36.26 KB | Columns: 13

Configure details

Create a time series analysis?

Enable this option to predict future activity over a specified date/time range. Data must be structured and sequential. [Learn more](#)

Yes

No

Training Screenshots

4 Selection of Prediction Parameters and Setting Experiment Settings

The screenshot displays the IBM Cloud Pak for Data interface for configuring an AutoAI experiment. The top navigation bar includes the IBM Cloud Pak for Data logo, a search bar, an upgrade button, a help icon, a notification bell with a red '1', the user's name 'Rahul Karunya Garige's Acc...', a dropdown menu, the location 'Frankfurt', and a user profile icon 'RG'.

The breadcrumb trail shows the path: [Projects](#) / [Loans Application](#) / Loans Application. The main title is 'Configure AutoAI experiment Loans Application', with an edit icon and an 'Autosaved: 15:44:51' timestamp.

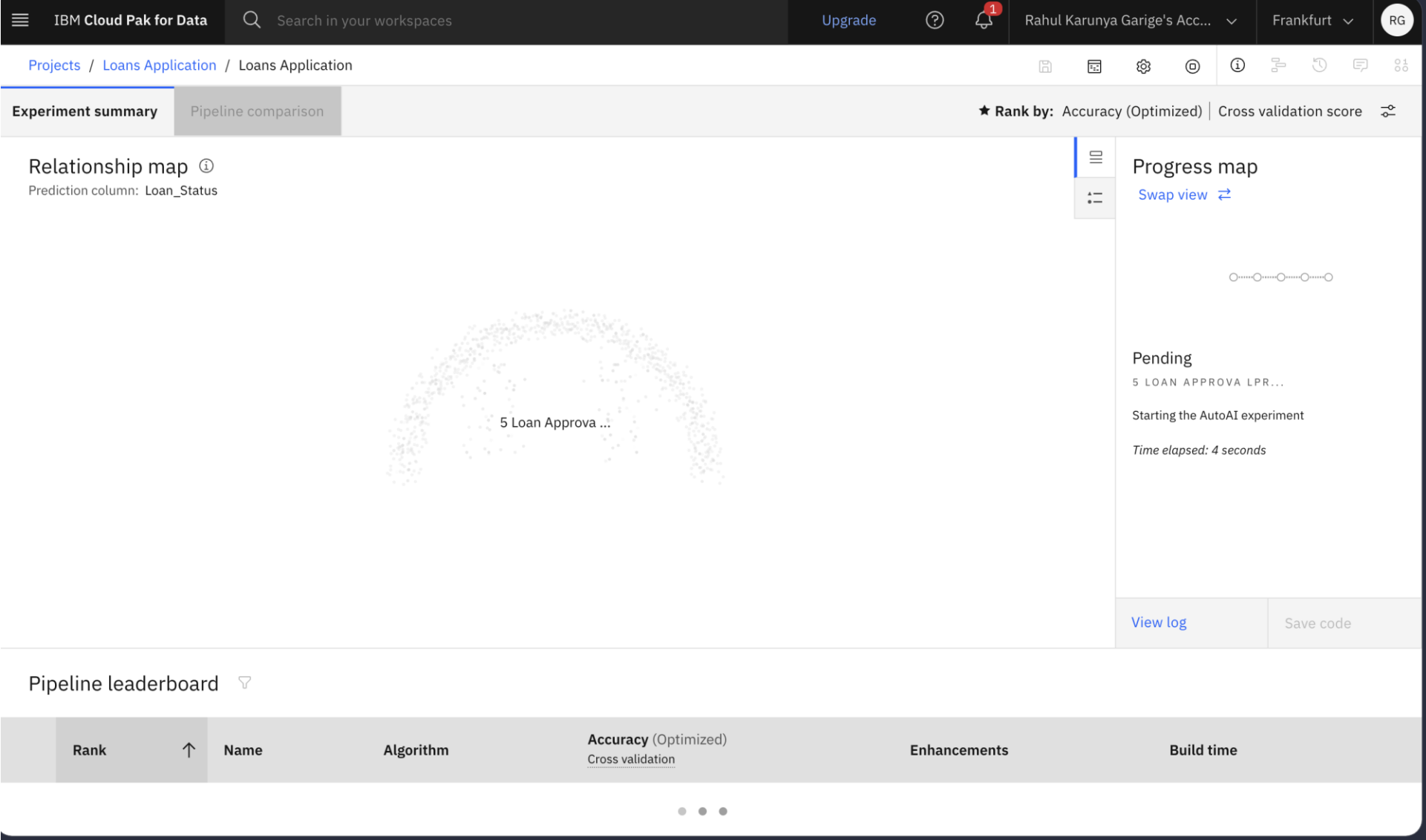
The interface is divided into two main panels:

- Add data source:** This panel contains a dashed box with the text 'Add files such as tabular data (CSV)'. Below this are two buttons: 'Browse' and 'Select from project'. A file named '5 Loan Approva lPrediction.csv' is listed with a size of 36.26 KB and 13 columns.
- Configure details:** This panel contains several sections:
 - Create a time series analysis?** A question with a line graph icon. The text below says 'Enable this option to predict future activity over a specified date/time range. Data must be structured and sequential. [Learn more](#)'. There are 'Yes' and 'No' buttons.
 - What do you want to predict?** A question with a lightbulb icon. Below it is a dropdown menu for 'Prediction column' with 'Loan_Status' selected.
 - Prediction column:** A section showing 'Loan_Status' and 'CUH remaining: 17.48 CUH'.
 - Prediction Type:** A section showing 'Binary Classification' and 'POSITIVE CLASS Y'. A note indicates it is 'OPTIMIZED FOR Accuracy & run time'.

At the bottom, there is a dark bar with 'Experiment settings' and a gear icon, and a blue bar with 'Run experiment' and a play button icon.

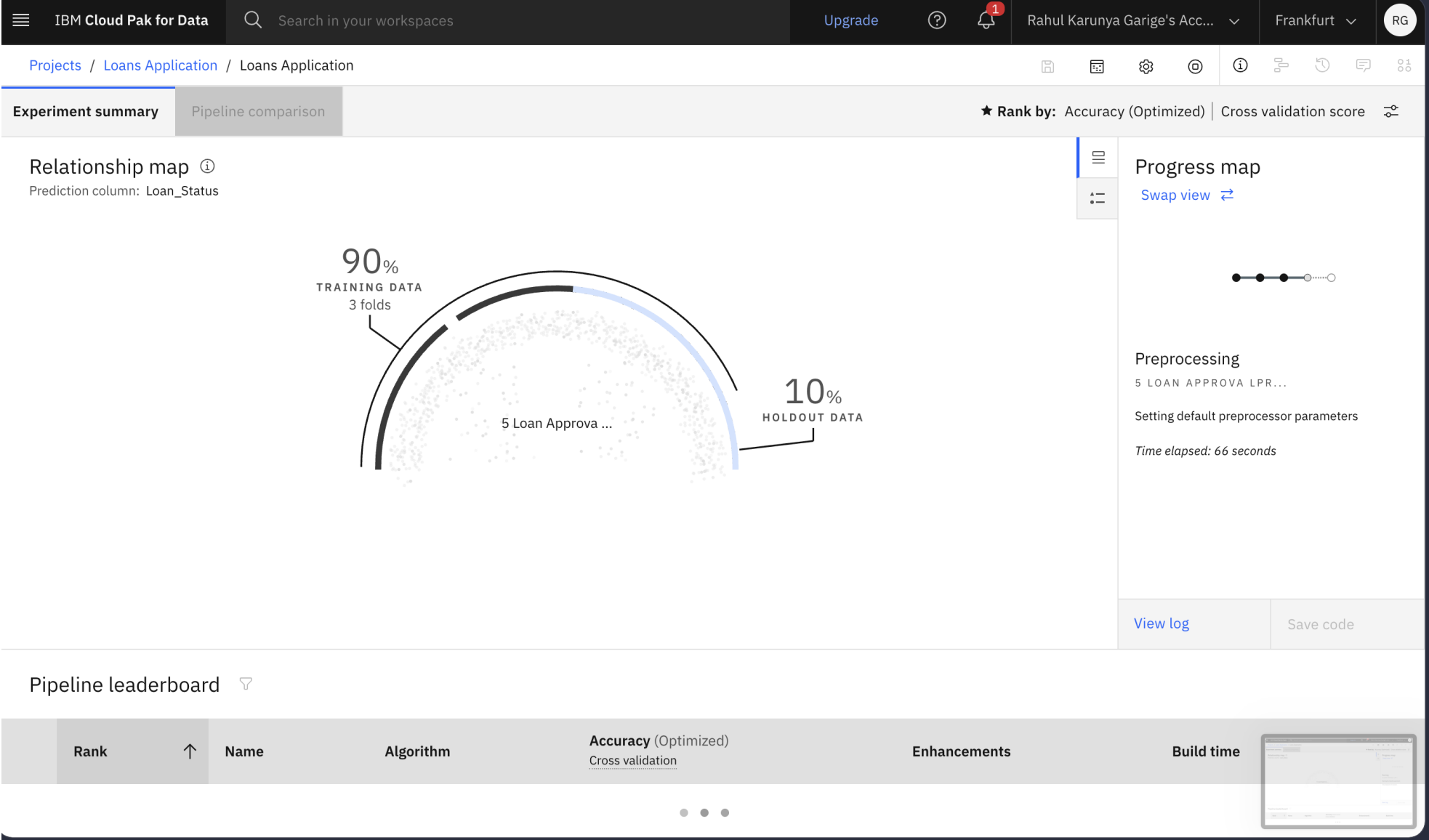
Training Screenshots

5 Experiment Running



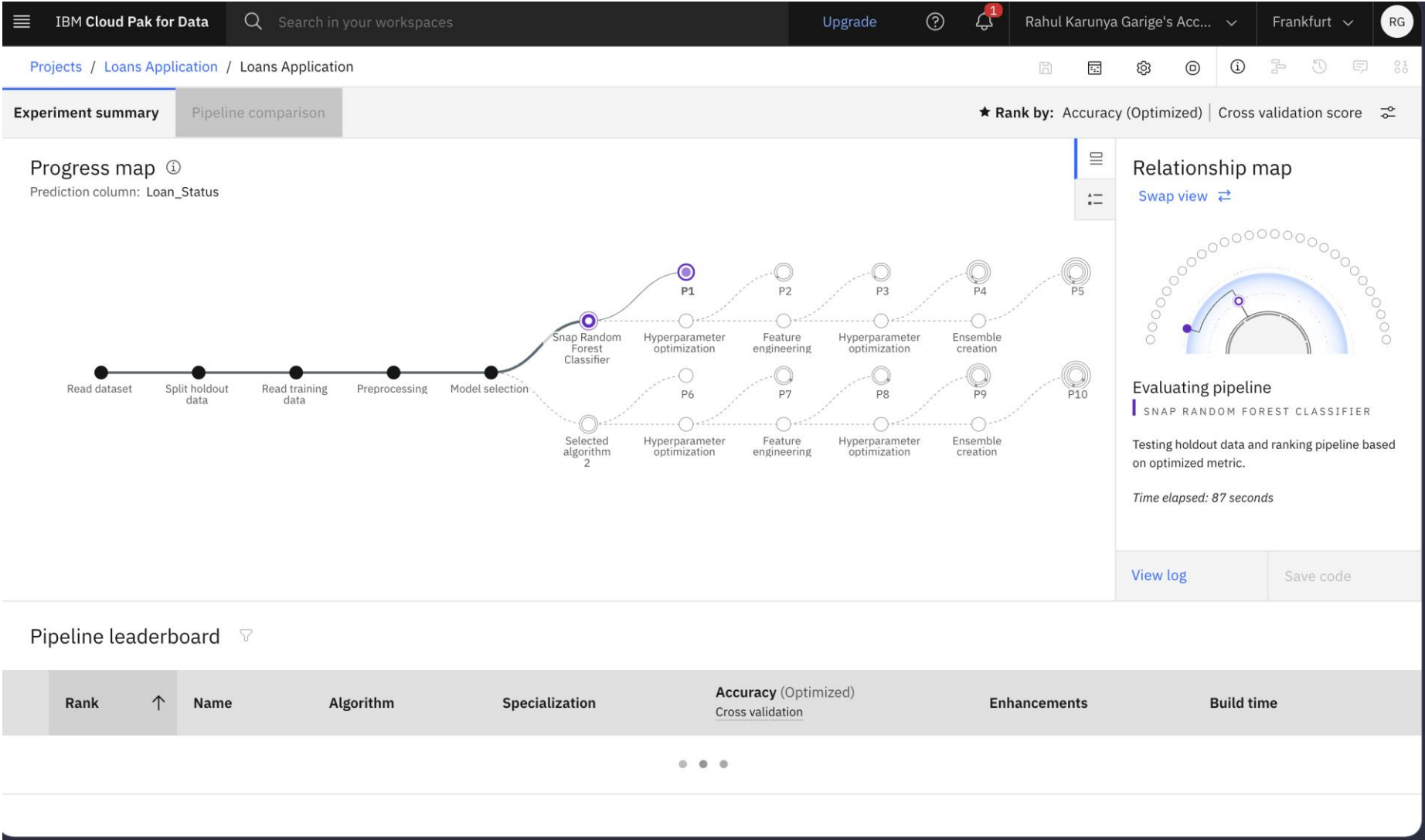
Training Screenshots

6 Experiment Running (Relationship Map)



Training Screenshots

7 Experiment Running (Progress Map)

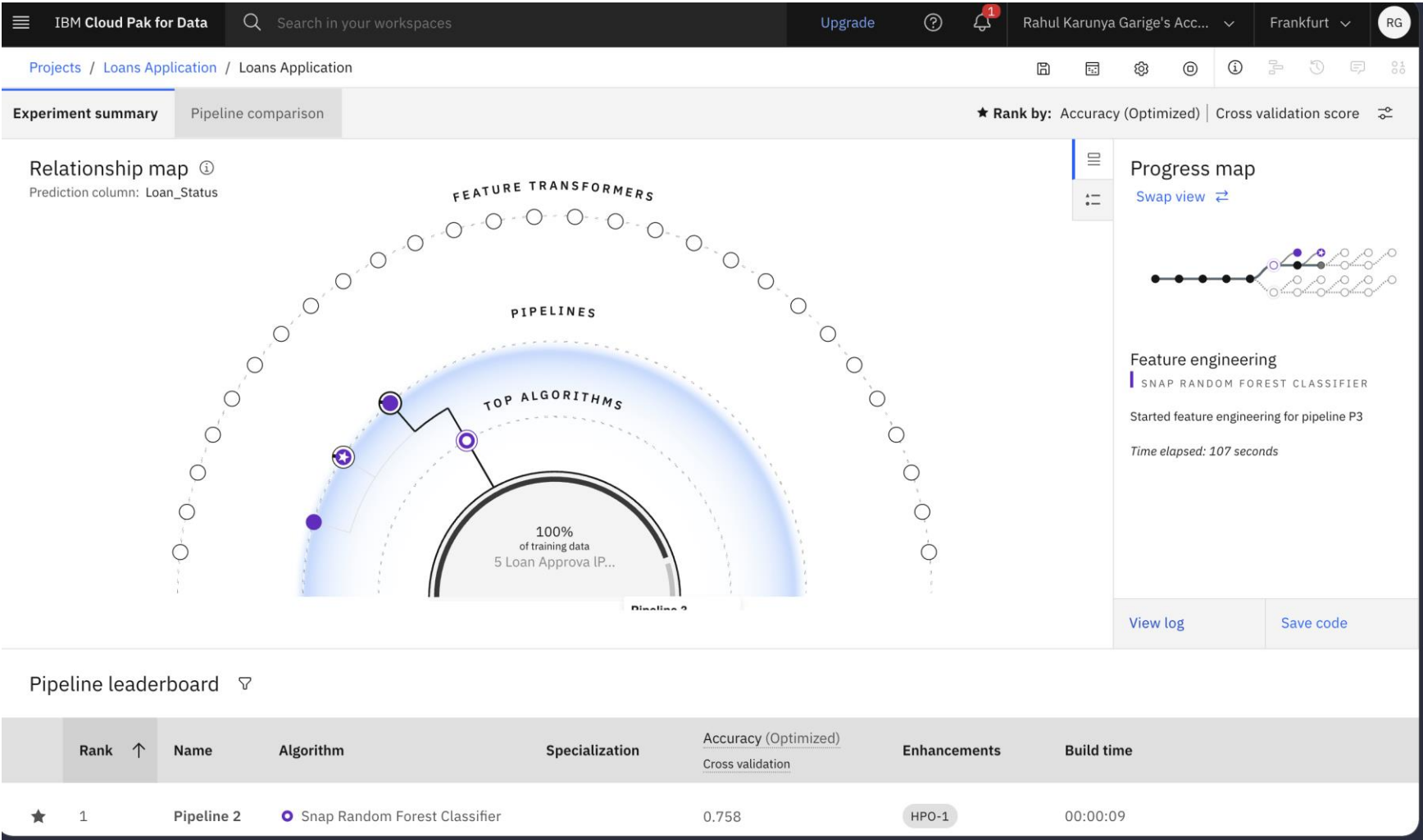


Pipeline leaderboard

Rank	↑	Name	Algorithm	Specialization	Accuracy (Optimized) Cross validation	Enhancements	Build time
...							

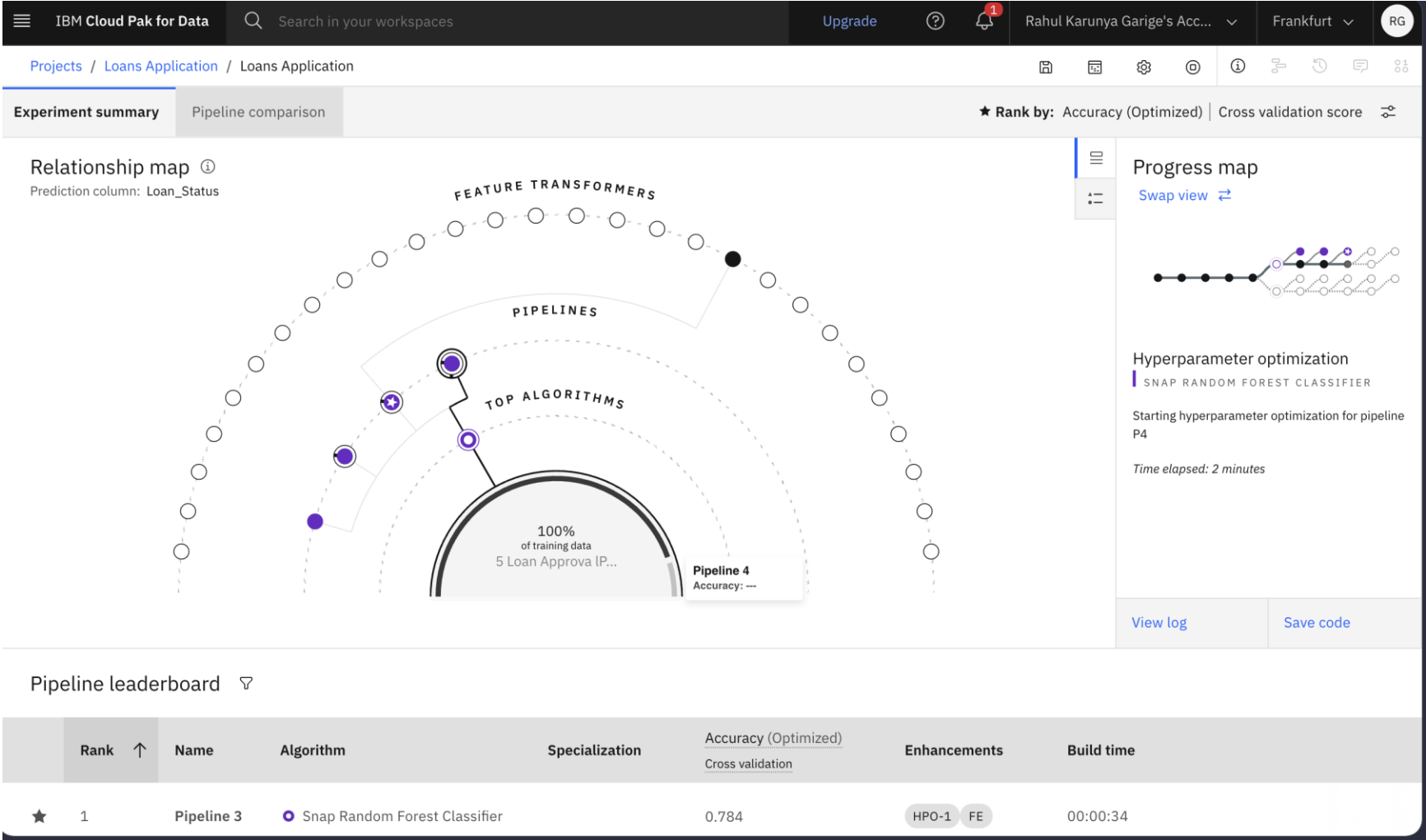
Training Screenshots

8 Experiment Running (Relationship Map Progress)



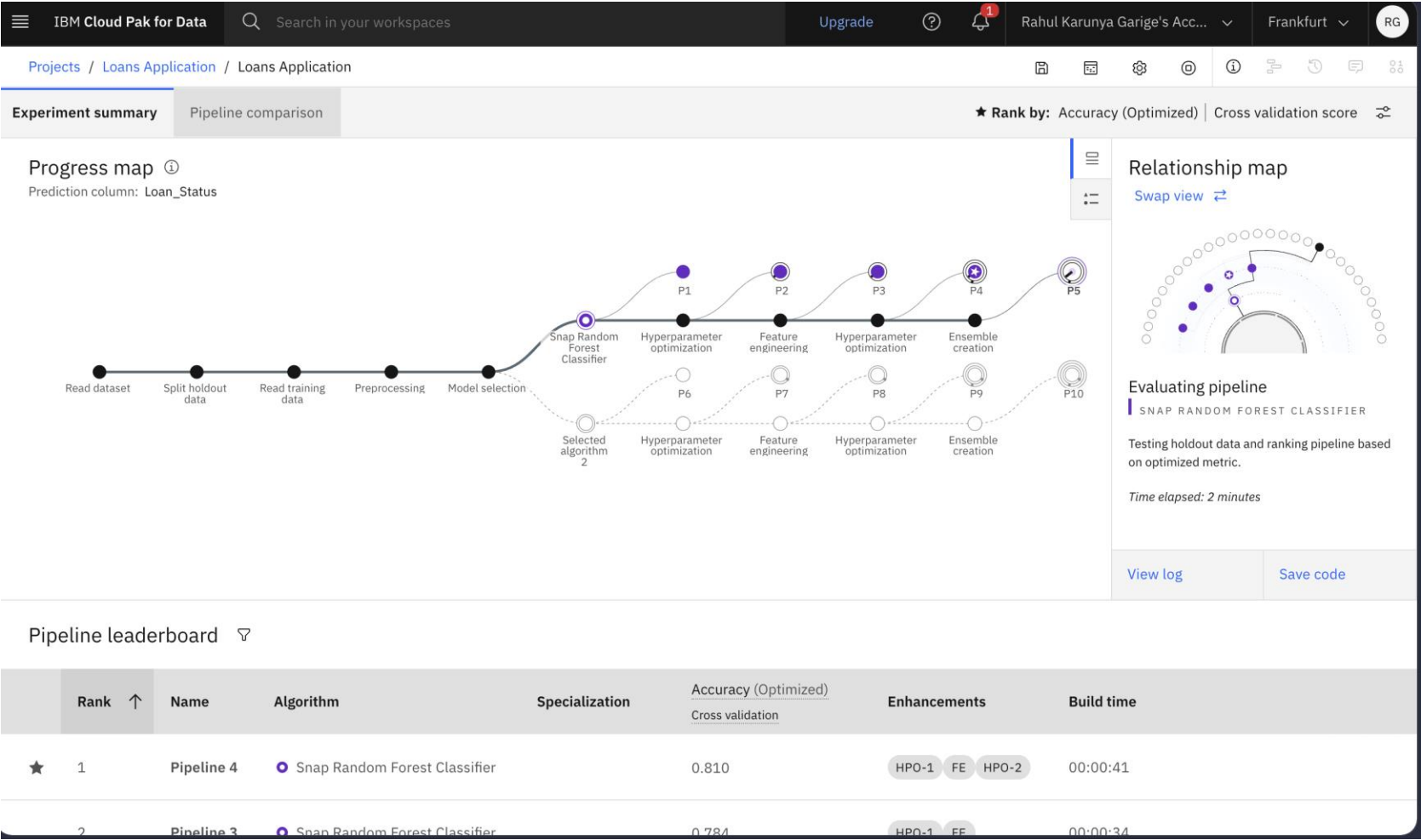
Training Screenshots

9 Experiment Running (Progress Map Progress)



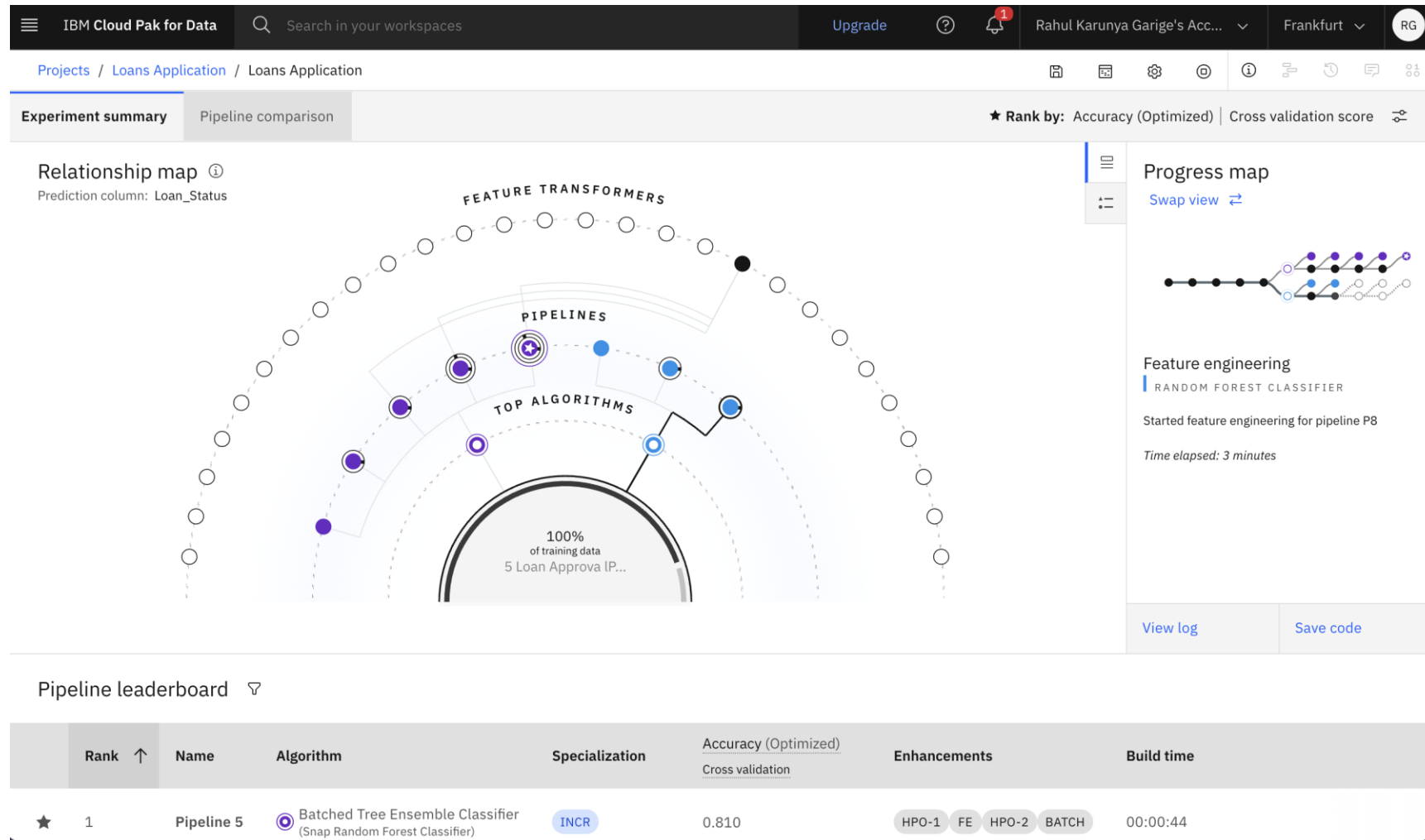
Training Screenshots

10 Experiment Running (Progress Map Progress)



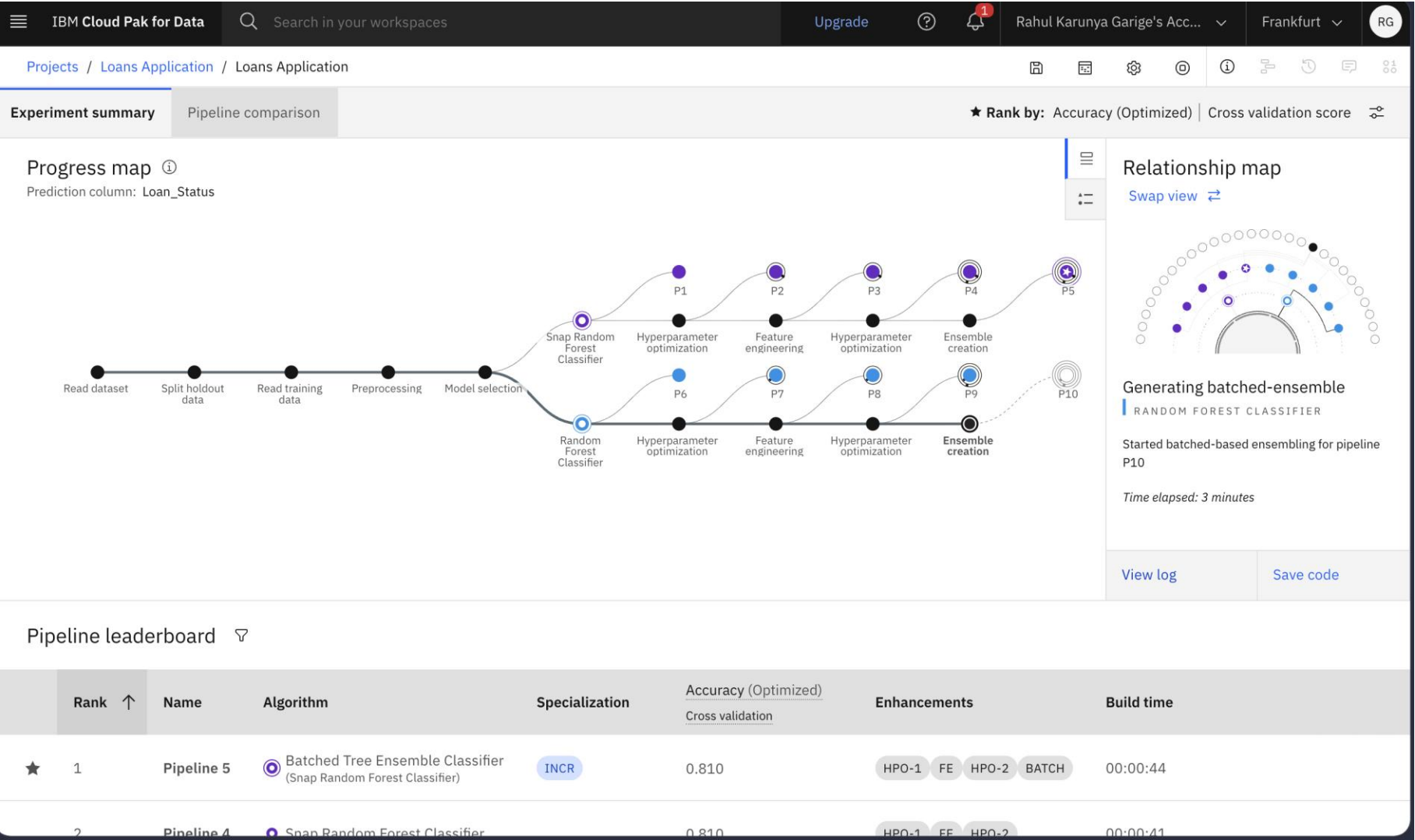
Training Screenshots

11 Experiment Running (Relationship Map Progress)



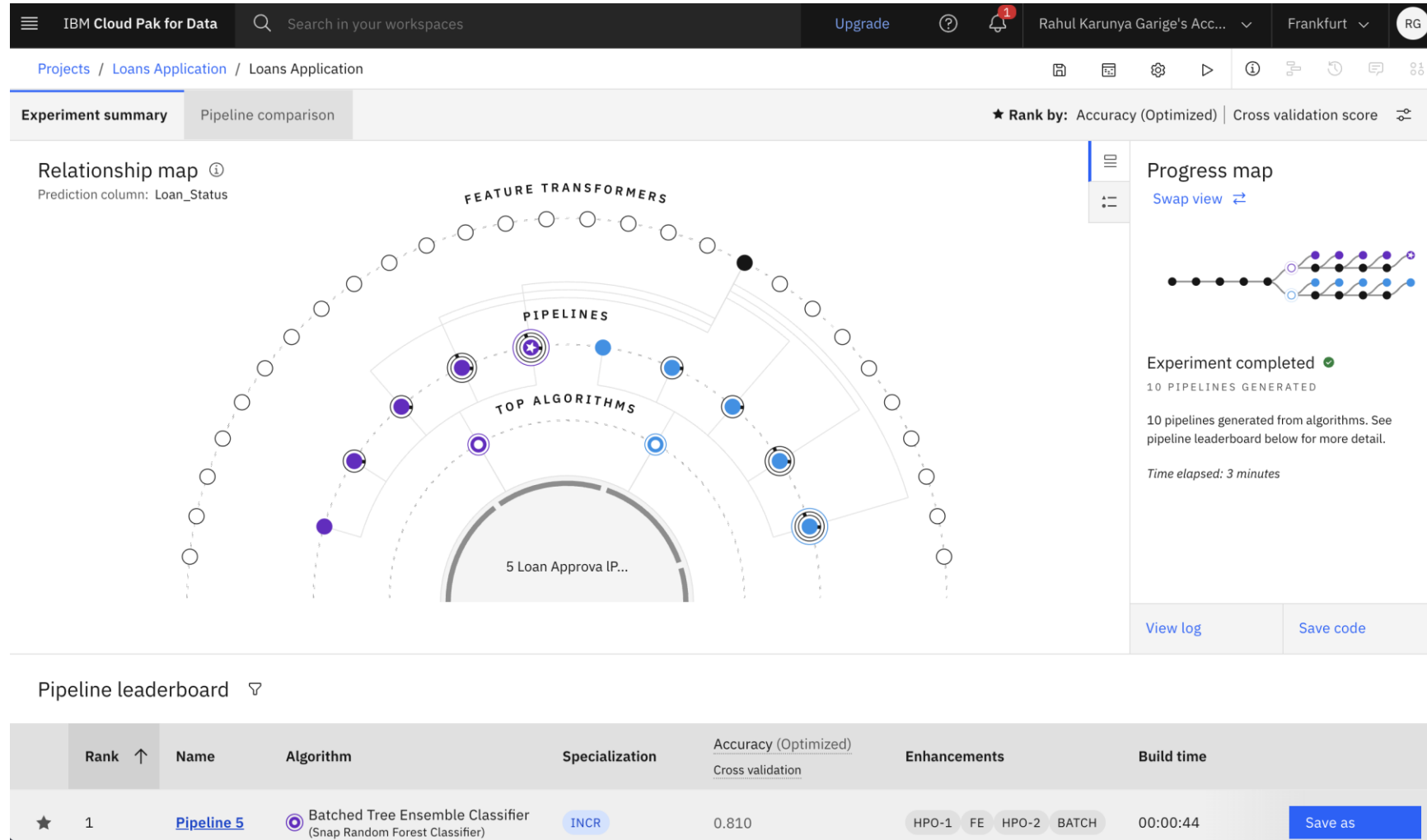
Training Screenshots

12 Experiment Running (Progress Map Progress)



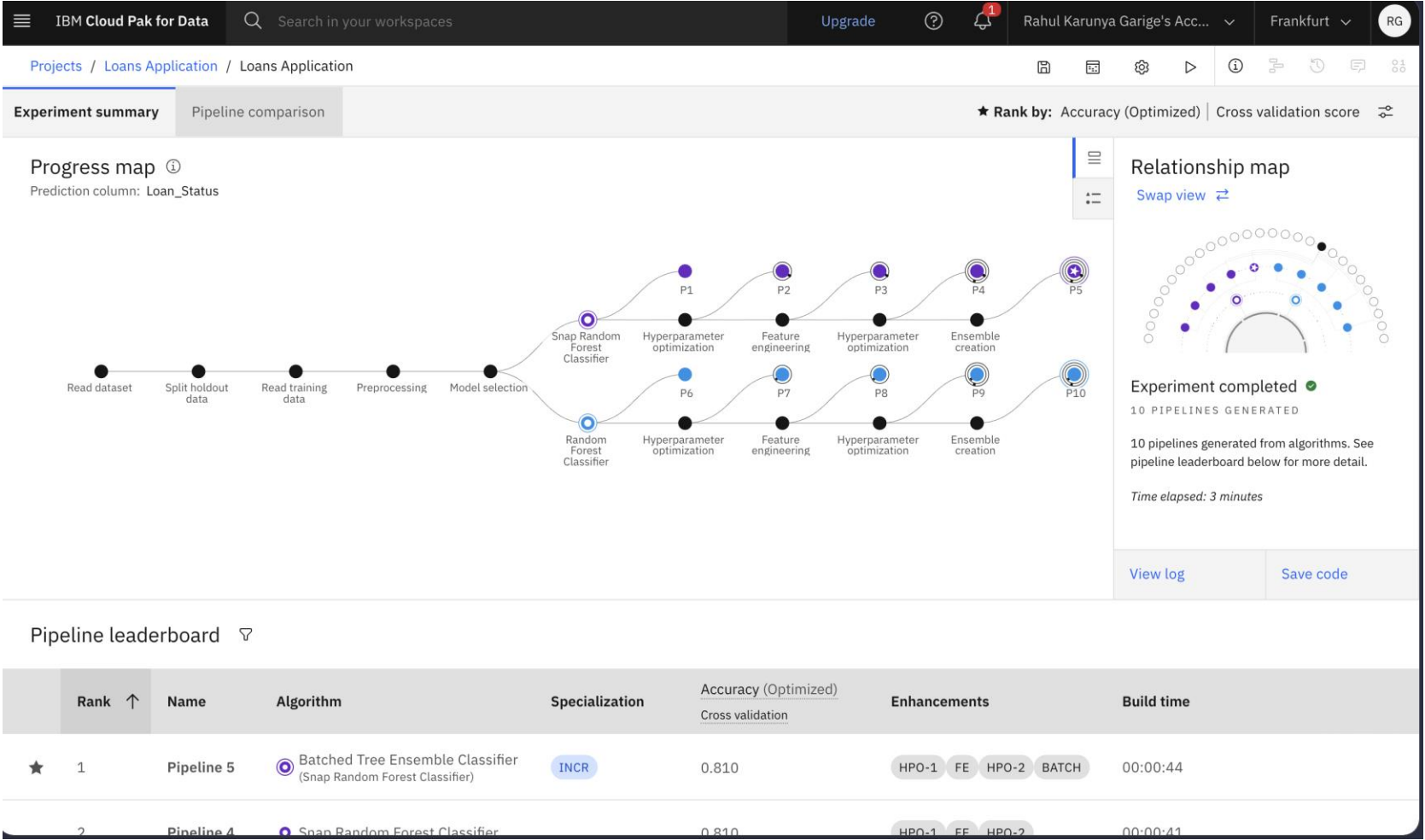
Training Screenshots

13 Experiment Completed(Relationship Map Progress)



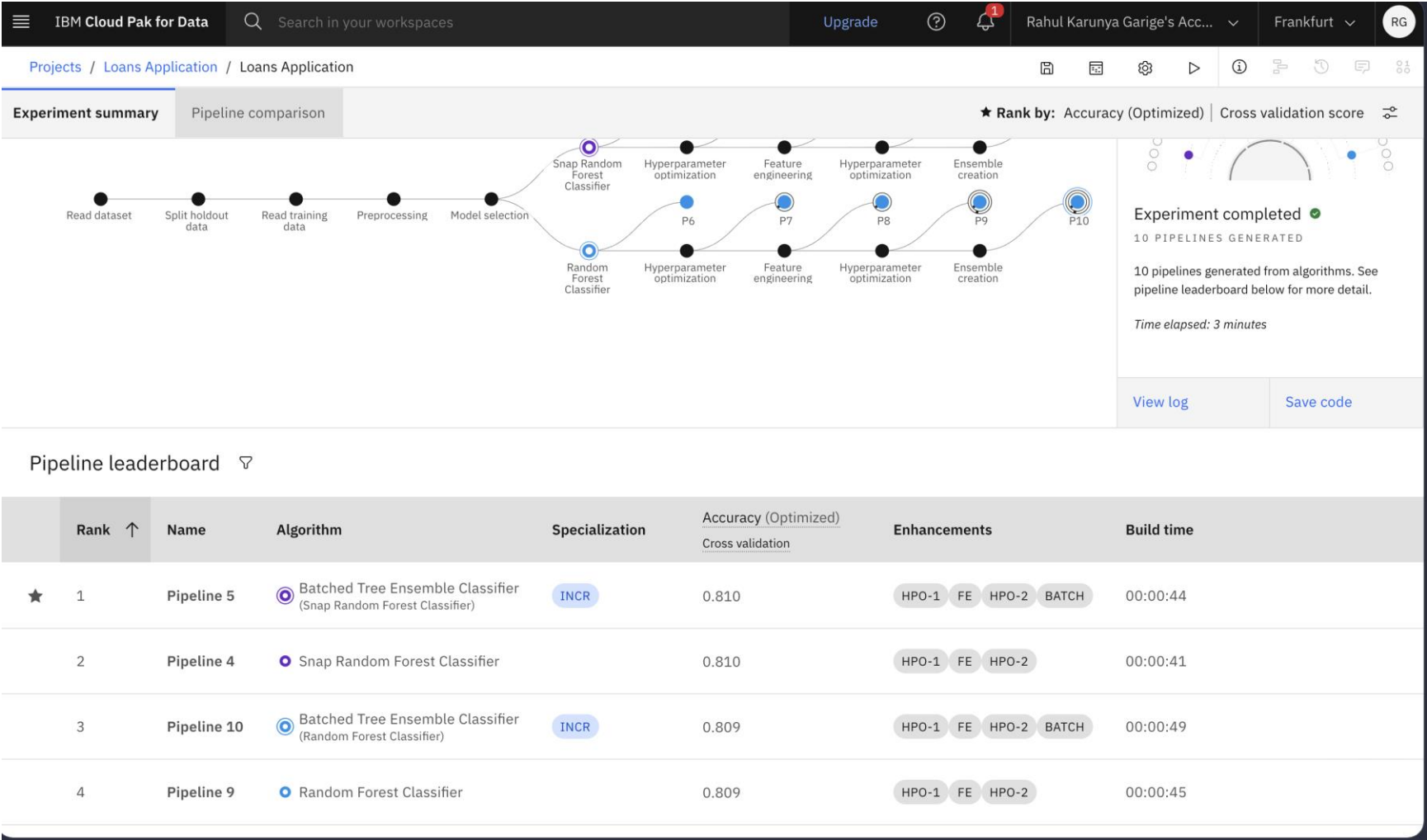
Training Screenshots

14 Experiment Completed(Progress Map Progress)



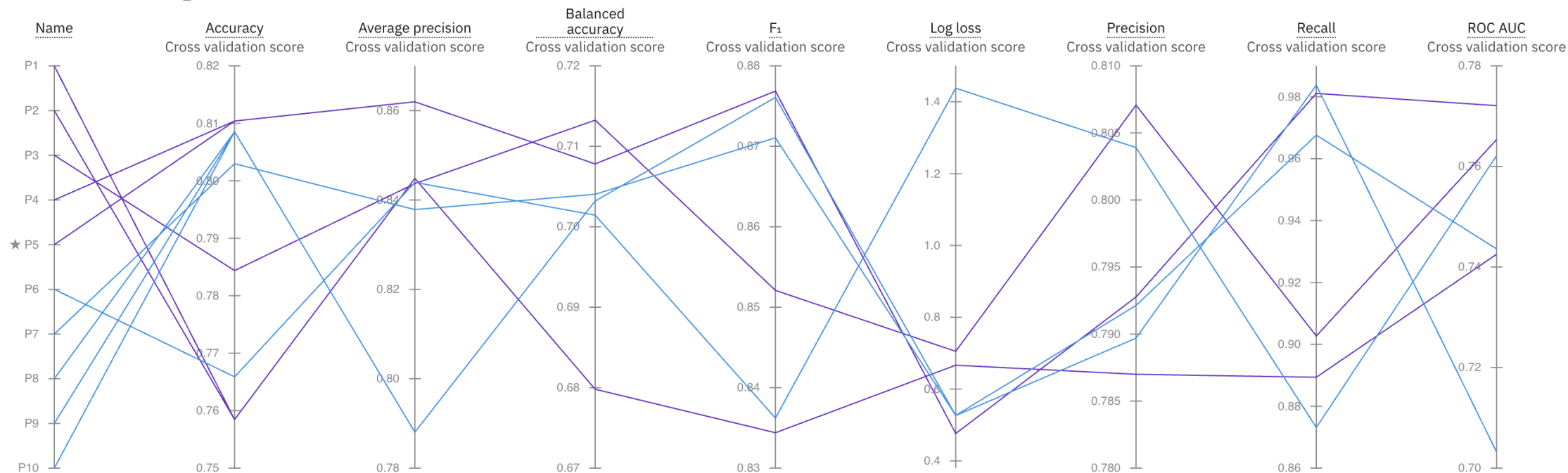
Training Screenshots

15 Experiment Completed (Various Pipelines)



Metric chart ⓘ

Prediction column: Loan_Status



Training Screenshots

Experiment Completed (Various Pipelines)

Training Screenshots

16 Model Saved

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

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Projects / Loans Application / Loans Application

Save as

Select asset type

Model ✓

Create a watsonx.ai Runtime model asset that you can test with new data, deploy to generate predictions, and trace lineage activity.

Notebook

Create a notebook if you want to view the code that created this model pipeline or interact with with the model programatically.

Define details

Name

P5 - Snap Random Forest Classifier: Loans Application

Description (optional)

Model description

Tags

Add tags to make assets easier to find.

Add a tag

Cancel

Create

Training Screenshots

17 Promote to Space

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

1

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Projects / Loans Application / P5 - Snap Random Forest Classifier: Loans Application

Promote to space

Input (1)

Column	Type
ApplicantIncome	integer
CoapplicantIncome	double
Credit_History	double
Dependents	double
Education	other
Gender	other
Loan_Amount_Term	double
Loan_ID	other

About this asset

Name

P5 - Snap Random Forest Classifier: Loans Application

Description

No description provided.

Asset Details

Type: wml-hybrid_0.1

Model ID: 019f0dbe-bb2e-77...

Software specification: hybrid_0.1

Hybrid pipeline software specifications: autoai-kb_rt25.1-py3.12

Tags

Add tags to make assets easier to find.

Last modified

7 seconds ago by Rahul Karunya Garige

Created on

Jun 28, 2026 by Rahul Karunya Garige

Training Screenshots

18 Creating Deployment Space

The screenshot shows the IBM Cloud Pak for Data interface. The top navigation bar includes the IBM Cloud Pak for Data logo, a search bar, an 'Upgrade' button, a help icon, a notification bell with a red '1', the user's name 'Rahul Karunya Garige's Ac...', the location 'Frankfurt', and a profile icon 'RG'. The breadcrumb trail is 'Projects / Loans Application / P5 - Snap Random Forest Classifier: Loans Application'. The main content area is titled 'Promote to space' with a subtitle 'Promote the asset to a deployment space to deploy the asset or to support a deployment'. A modal dialog box titled 'Create a deployment space' is open, with a subtitle 'Use a space to collect assets in one place to create, run, and manage deployments'. The dialog has two tabs: '+ New' (selected) and 'Local file'. The 'Define details' section contains a 'Name' field with the value 'Project', a 'Description (Optional)' field with the placeholder text 'What's the purpose of this space?' and a character count '0/100', and a 'Deployment stage' dropdown menu with the text 'Select or enter a name that describes the purpose of the space'. At the bottom of the dialog are 'Cancel' and 'Create' buttons.

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

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Projects / Loans Application / P5 - Snap Random Forest Classifier: Loans Application

Promote to space

Promote the asset to a deployment space to deploy the asset or to support a deployment

Create a deployment space

Use a space to collect assets in one place to create, run, and manage deployments

+ New

Local file

Define details

Name

Project

Description (Optional) 0/100

What's the purpose of this space?

Deployment stage ⓘ

Select or enter a name that describes the purpose of the space

Cancel

Create

Training Screenshots

19 Promote to Space

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

?

1

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Frankfurt ▾

RG

Projects / Loans Application / P5 - Snap Random Forest Classifier: Loans Application

Input

Columns

Applications

Cooperative

Credit

Dependencies

Education

Generation

Loan

Loan

Promote to space

Promote the asset to a deployment space to deploy the asset or to support a deployment.

Target deployment space

Project ▾

Why don't I see all of my spaces? ⓘ

☐ Go to the model in the space after promoting it

Description (Optional)

Description of assets

Find or create tags ▾

ⓘ Promoting a version of an asset to a space creates a new asset in the space, with a new asset ID.

Selected assets (1)

Name	Format	Version	Status
P5 - Snap Random Forest Classifier...	Model	Cur... ▾	Queued

Promoting an asset promotes dependent assets as well. For example, promoting a model also promotes the associated software specification and package extensions. You will see all promoted assets in the target space.

Cancel

Promote

Training Screenshots

20 Promote to Space Successful

The screenshot displays the IBM Cloud Pak for Data interface. The top navigation bar includes the IBM Cloud Pak for Data logo, a search bar, an 'Upgrade' button, a help icon, a notification bell with a red '1', the user's name 'Rahul Karunya Garige's Ac...', the location 'Frankfurt', and a profile icon 'RG'.

The breadcrumb trail shows the path: **Projects / Loans Application / P5 - Snap Random Forest Classifier: Loans Application**.

The main content area is titled 'Promote to space' with the subtitle 'Promote the asset to a deployment space to deploy the asset or to support a deployment.' Below this, a green checkmark icon and the text 'Promotion completed.' indicate success.

A table titled 'Selected assets (1)' lists the promoted asset:

Name	Format	Version	Status
P5 - Snap Random Forest Classifier: Loans Ap...	Model	Current	Promoted ✓

Below the table, a note states: 'Promoting an asset promotes dependent assets as well. For example, promoting a model also promotes the associated software specification and package extensions. You will see all promoted assets in the target space.'

A green success toast notification is visible in the top right corner, stating: 'Success: Successfully promoted P5 - Snap Random Forest Classifier: Loans Application to the deployment space. Go to the deployment space to prepare the assets for deployment. Timestamp 3:52:08 PM'.

A blue 'Close' button is located at the bottom right of the modal.

Training Screenshots

21 Deployment Spaces

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

1

Rahul Karunya Garige's Ac...

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Deployment spaces /

Project

OverviewAssetsDeploymentsJobsManage

Space

General

Access control

Environments

Resource usage

Details

Name

Project

Description

No description provided.

Space GUID

09516691-7d77-4177-97db-92a5e7c2a51b

Date created

Jun 28, 2026, 3:51 PM

by Rahul Karunya Garige (You)

Last updated

Jun 28, 2026, 3:52 PM

Stage

Not provided

Stage type

Pre-production

Tags

No tags are set to this space.

Controls

Storage

Storage used

11.08 KB

Name

CloudObjectStorage

Bucket

0a702b5b-b140-409a-87ca-c0201e7256ee

Regenerate credentials

Last updated on: Jun 28, 2026, 03:51 PM

Regenerate

Manage in IBM Cloud

watsonx.ai Runtime service

WatsonMachineLearning

Training Screenshots

22 Create Deployment

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

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Deployment spaces / Project / P5 - Snap Random Forest Classifier: Loans Application

Create a deployment

Define details

Associated asset
P5 - Snap Random Forest Classifier: Loans Application

Deployment type

Online ☒

Run the model on data in real-time, as data is received by a web service.

Batch

Run the model against data as a batch process.

Name

Loans Application

Serving name

Deployment serving name

Enter a short name to be used as the serving name for the deployment. The name must be unique to be valid.

Cancel Create

Training Screenshots

23 Deployment

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

1

Rahul Karunya Garige's Ac...

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Deployment spaces / Project / P5 - Snap Random Forest Classifier: Loans Application

Deployments

Model details

Search

New deployment

Name	Type	Status	Tags	Last modified	
 Loans Application	Online	 Initializing		4 seconds ago Rahul Karunya Garige (You)	

Items per page: 20

1-1 of 1 items

1 of 1 pages

About this asset

Name

P5 - Snap Random Forest Classifier: Loans Application

Description

No description provided.

Asset Details

Type: wml-hybrid_0.1

Model ID: 019f0dc0-2909-74...

Software specification: hybrid_0.1

Hybrid pipeline software specifications: autoai-kb_rt25.1-py3.12

Tags

Add tags to make assets easier to find.

Source asset details

Last modified
1 minute ago by Rahul Karunya Garige

Created on
Jun 28, 2026 by Rahul Karunya Garige

Training Screenshots

24 Deployment Completed

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

2

Rahul Karunya Garige's Ac...

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Deployment spaces / Project / P5 - Snap Random Forest Classifier: Loans Application

Deployments

Model details

Search

New deployment

Name	Type	Status	Tags	Last modified	
Loans Application	Online	Deployed		37 seconds ago Rahul Karunya Garige (You)	

Items per page: 20

1-1 of 1 items

1 of 1 pages

About this asset

Name

P5 - Snap Random Forest Classifier: Loans Application

Description

No description provided.

Asset Details

Type: wml-hybrid_0.1

Model ID: 019f0dc0-2909-74...

Software specification:
hybrid_0.1

Hybrid pipeline software specifications:
autoai-kb_rt25.1-py3.12

Tags

Add tags to make assets easier to find.

Source asset details

Last modified
1 minute ago by Rahul Karunya Garige

Created on
Jun 28, 2026 by Rahul Karunya Garige

25 Navigation to the Project

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

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Deployment spaces / Project / P5 - Snap Random Forest Classifier: Loans Appli... /

Loans Application

Deployed Online

API reference

Test

Evaluations

Transactions

Endpoints for scoring ⓘ

Private endpoint

https://private.eu-de.ml.cloud.ibm.com/ml/v4/deployments/019f0dc1-94fa-71a4-9c77-d7a300b3 ⓘ

Bearer <token> ⓘ

IAM

Public endpoint

https://eu-de.ml.cloud.ibm.com/ml/v4/deployments/019f0dc1-94fa-71a4-9c77-d7a300b348e9/pre ⓘ

[Learn more](#) about the 2021-05-01 version query parameter

Code snippets

cURL

Java

JavaScript

Python

Scala

NOTE: you must set \$API_KEY below using information retrieved from your IBM Cloud account (https://eu-de.dataplatform.cl ⓘ
export API_KEY=<your API key>

export IAM_TOKEN=\$(curl --insecure -X POST --location "https://iam.cloud.ibm.com/identity/token" --header "Content-Type: appli

TODO: manually define and pass values to be scored below

curl --location "https://private.eu-de.ml.cloud.ibm.com/ml/v4/deployments/019f0dc1-94fa-71a4-9c77-d7a300b348e9/predictions?ver
 {

About this deployment ×

Name ⓘ
Loans Application

Description ⓘ
No description provided.

Deployment Details

Deployment ID: 019f0dc1-94fa-71... ⓘ
Serving name:
No serving name
Software specification: ⓘ
hybrid_0.1 ⓘ
Hybrid pipeline software specifications:
autoai-kb_rt25.1-py3.12
Copies: ⓘ
1

Tags ⓘ
Add tags to make assets easier to find.

Associated asset ⓘ

🔗 P5 - Snap Random Forest Classifier: Lo...
019f0dc0-2909-7422-86fd-aa9fc84b8c80

Last modified
1 minute ago

Created on
Jun 28, 2026

Training Screenshots

26 Test Data 1

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

2

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Deployment spaces / Project / P5 - Snap Random Forest Classifier: Loans Appli... /

Loans Application ✔ Deployed Online

API referenceTestEvaluationsTransactions

Enter input data

TextJSON

Enter data manually or use a CSV file to populate the spreadsheet. Max file size is 50 MB.

:

Clear all ×

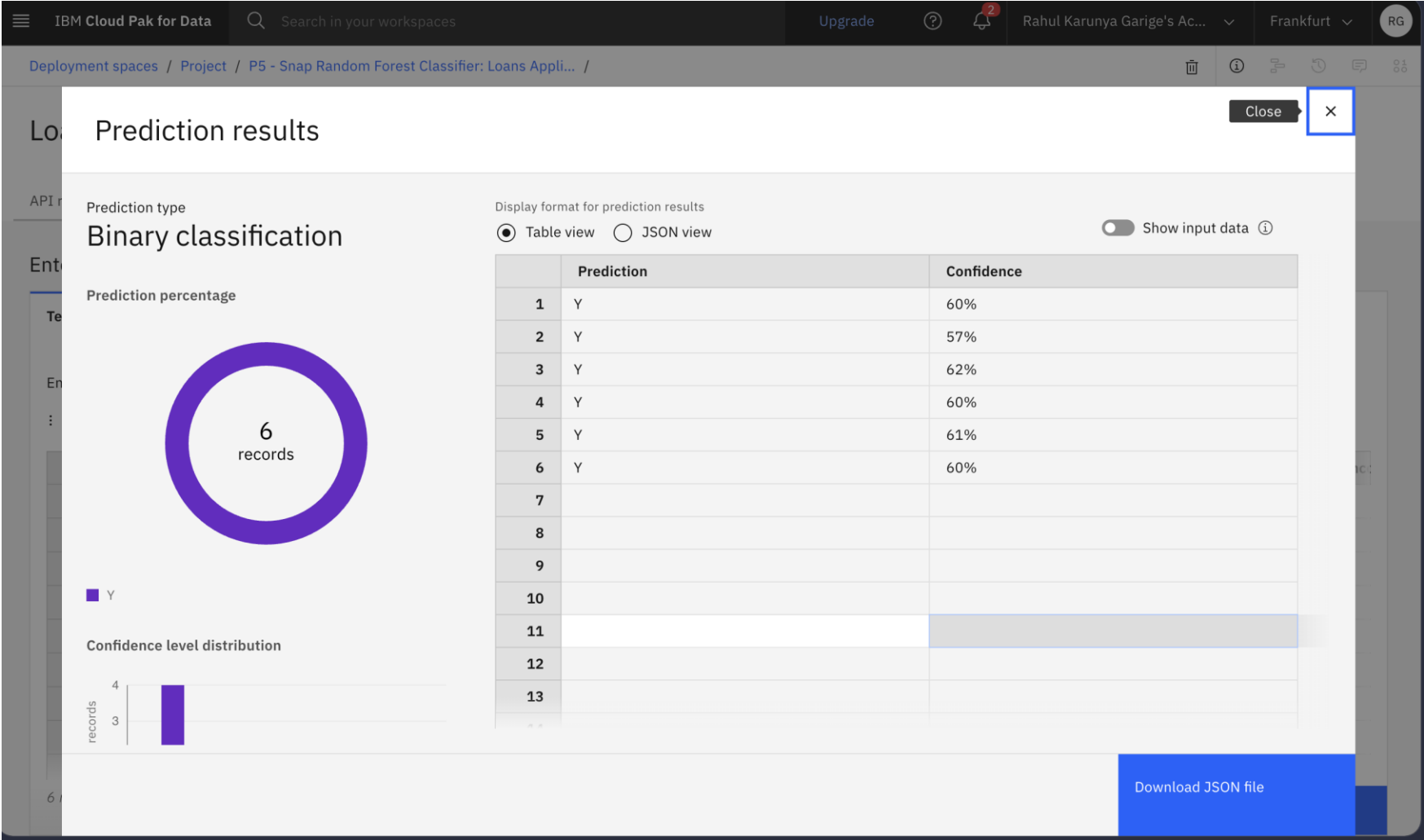
	Loan_ID (other)	Gender (other)	Married (other)	Dependents (double)	Education (other)	Self_Employed (other)	ApplicantIncome (integer)	CoapplicantIncome (integer)
1	TEST_01	Female	Yes	2.0	Graduate	Yes	50000	10000
2	TEST_02	Male	No	0.0	Graduate	No	29000	0.0
3	TEST_03	Male	Yes	4.0	Graduate	No	100000	25000
4	TEST_04	Female	No	1.0	Not Graduate	No	90000	0.0
5	TEST_05	Male	Yes	2.0	Graduate	No	56000	10000
6	TEST_06	Female	Yes	0.0	Not Graduate	Yes	64000	49000
7								
8								
9								

6 rows, 12 columns

Predict

Training Screenshots

27 Test Data 1 Prediction Result



Training Screenshots

28 Test Data Set 2

IBM Cloud Pak for Data

Search in your workspaces

Upgrade

2

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Deployment spaces / Project / P5 - Snap Random Forest Classifier: Loans Appli... /

Loans Application Deployed Online

API referenceTestEvaluationsTransactions

Enter input data

Text

JSON

Enter data manually or use a CSV file to populate the spreadsheet. Max file size is 50 MB.

:

Clear all

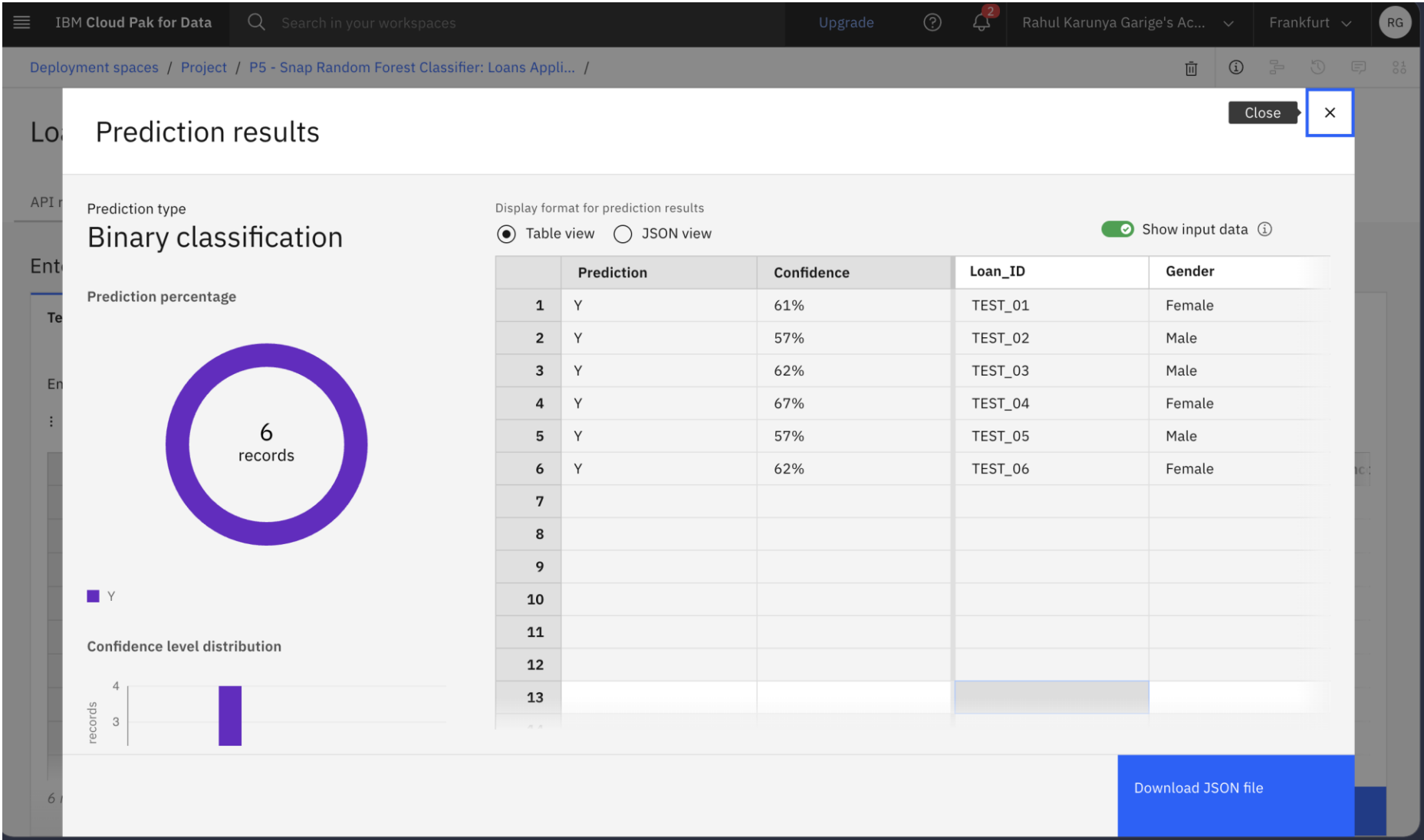
	Loan_ID (other)	Gender (other)	Married (other)	Dependents (double)	Education (other)	Self_Employed (other)	ApplicantIncome (integer)	CoapplicantIncome (integer)
1	TEST_01	Female	Yes	2	Graduate	Yes	44000	0
2	TEST_02	Male	Yes	6	Graduate	No	29000	0
3	TEST_03	Male	Yes	4	Graduate	No	67000	25000
4	TEST_04	Female	Yes	1	Not Graduate	No	9000	0
5	TEST_05	Male	Yes	2	Graduate	No	34000	5000
6	TEST_06	Female	Yes	10	Not Graduate	Yes	64000	49000
7								
8								
9								

6 rows, 12 columns

Predict

Training Screenshots

29 Test Data Set 2 Prediction



Close

X

Training Screenshots

Website Building Using IBM Bob

The screenshot displays the IBM Bob IDE interface for a project named "loan-approval-webapp". The Explorer panel on the left shows the project structure, including files like `__init__.py`, `auth.py`, `predictor.py`, `utils.py`, `static`, `css`, `images`, `js`, and `templates`. The `templates` directory contains `about.html`, `base.html`, `error.html`, `index.html`, `predict.html`, and `result.html`. The Outline panel shows the structure of the `index.html` file, including sections like `section.hero-section`, `div.hero-overlay`, `div.container.position-relative.z-1`, `div.row.align-items-center.min-vh-h...`, and `div.col-lg-7`.

The main editor displays the `index.html` file, which is a Jinja2 template. It includes a hero section with a title "Instant Loan Approval" and a subtitle "Get an intelligent, data-driven loan eligibility decision in seconds. Our system uses a production-grade AutoAI model deployed on IBM Cloud to analyze your financial profile." The template also features a "Check Eligibility Now" button and a "How It Works" button. The hero stats section displays three statistics: "Model Accuracy" (98%), "Response Time" (<2s), and "Input Features" (12).

The right sidebar shows the "IBM BOB" status, indicating that the "Server is LIVE". It also displays a "Pasted text #1" snippet and a "Todo List" with 19 items, including tasks like "Create services/auth.py (IBM IAM to...)", "Create services/predictor.py (IBM A...)", "Create services/utlis.py (helper utilit...)", "Create app.py (Flask application)", "Create templates/base.html", "Create templates/index.html (Home ...)", "Create templates/predict.html (Pred...)", "Create templates/result.html (Result...)", "Create templates/about.html (About...)", "Create templates/error.html (Error p...", "Create static/css/style.css", "Create static/js/main.js", "Create README.md", and "Validate all files are complete and c...".

Training Screenshots

Website Building – API Key Generation

Search resources and products...

Q

Catalog

Manage ▾

Rahul Karunya Garige's Ac...

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API keys

Create, view, and work with API keys that you have access to manage. IBM Cloud API keys are associated with a user's identity and can be used to access cloud platform and classic infrastructure APIs, depending on the access that is assigned to the user. The following table displays a list of API keys created in this account. [Learn more](#).

Looking for more options to manage API Keys? Try [IBM Cloud® Secrets Manager](#) for creating and leasing API keys dynamically and storing them securely in your own dedicated instance.

Unused or overly permissive API keys increase the risk of unauthorized access. Regularly review the [Inactive identities report](#), rotate keys, and apply only the minimum required permissions.

API keys associated with a user's identity have the same access that the user is assigned across all accounts. To update the access for an API key, assign or remove access for the user.

View

My IBM Cloud API keys ▾

Q

Filter by API key name or description

Create +

ID	Name	↓	Description	Date created	Status	Enabled	Expires in
📄	cpd-apikey-IBMid-695001NQHO-2026-06-18T12:45:06Z		API key created/managed by task credentials. It is managed for your use with Watson Studio operations. Please do not delete here.	18-06-2026 12:45 GMT	🔒	☒ Yes	-
📄	loan			29-06-2026 15:53 GMT	🔒	☒ Yes	90 days

Training Screenshots

Website Working Using IBM API Reference

The screenshot displays the LoanAI website interface. At the top, there is a dark blue header with the 'LoanAI' logo on the left and navigation links for 'Home', 'Predict', and 'About' in the center. A prominent blue button labeled 'Get Prediction' is on the right. Below the header, a badge states 'Powered by IBM watsonx.ai AutoAI'. The main heading reads 'Instant Loan Approval Prediction with AI', with 'Prediction' in blue. A subtext explains: 'Get an intelligent, data-driven loan eligibility decision in seconds. Our system uses a production-grade AutoAI model deployed on IBM Cloud to analyze your financial profile.' Below this are two buttons: 'Check Eligibility Now' and 'How It Works'. At the bottom, three statistics are listed: '98% MODEL ACCURACY', '<2s RESPONSE TIME', and '12 INPUT FEATURES'. On the right side, a 'Loan Application' summary card shows input details: Applicant Income (₹ 5,000), Loan Amount (₹ 128,000), Credit History (Good), and Property Area (Urban). At the bottom of this card, a green box displays 'APPROVED' with a 92.4% confidence level.

LoanAI

Home Predict About Get Prediction

Powered by IBM watsonx.ai AutoAI

Instant Loan Approval Prediction with AI

Get an intelligent, data-driven loan eligibility decision in seconds. Our system uses a production-grade AutoAI model deployed on IBM Cloud to analyze your financial profile.

Check Eligibility Now How It Works

98% MODEL ACCURACY <2s RESPONSE TIME 12 INPUT FEATURES

Loan Application

Applicant Income	₹ 5,000
Loan Amount	₹ 128,000
Credit History	Good
Property Area	Urban

APPROVED 92.4% Confidence

Training Screenshots

Website Working Using IBM API Reference



[Home](#) [Predict](#) [About](#) [Get Prediction](#)

THE PROCESS

How It Works

From form submission to AI prediction in 3 steps



01

Fill the Form

Enter your personal and financial details in our secure prediction form.



02

AI Analysis

Our Flask backend calls the IBM watsonx.ai deployment API with your data.



03

Instant Result

Receive an Approved or Rejected decision with a confidence percentage.

AI PLATFORM

Powered by IBM watsonx.ai

IBM watsonx.ai is an enterprise-grade AI and data platform for business. Our loan model was trained using AutoAI — IBM's automated machine learning engine that evaluates dozens of algorithms to find the optimal model pipeline.

- AutoAI pipeline optimization
- Binary classification (Approved / Rejected)
- Online deployment on IBM Cloud eu-de
- REST API with IAM bearer-token auth

[→ Learn More](#)

**User**
Submits loan application form

↓

**Flask Backend**
Validates data, obtains IAM token

↓

**IBM watsonx.ai API**
eu-de.ml.cloud.ibm.com

↓

**Prediction Result**
Approved / Rejected + Confidence

Training Screenshots

Website Working – Loan Application Form with Test Data 1

[Home](#) [Predict](#) [About](#) [Get Prediction](#)

 **Loan Application Details**

 **PERSONAL INFORMATION**

Gender *
Male

Married *
Yes

Dependents *
3+

Education *
Graduate

Self Employed *
Yes

 **FINANCIAL INFORMATION**

Applicant Income Monthly income in USD/INR *
\$ 10000

Coapplicant Income Enter 0 if none *
\$ 5000

Loan Amount In thousands *
50000 K

Loan Amount Term *
120 months (10 yrs)

 **LOAN & PROPERTY DETAILS**

Credit History *
1 — Good (meets guidelines)

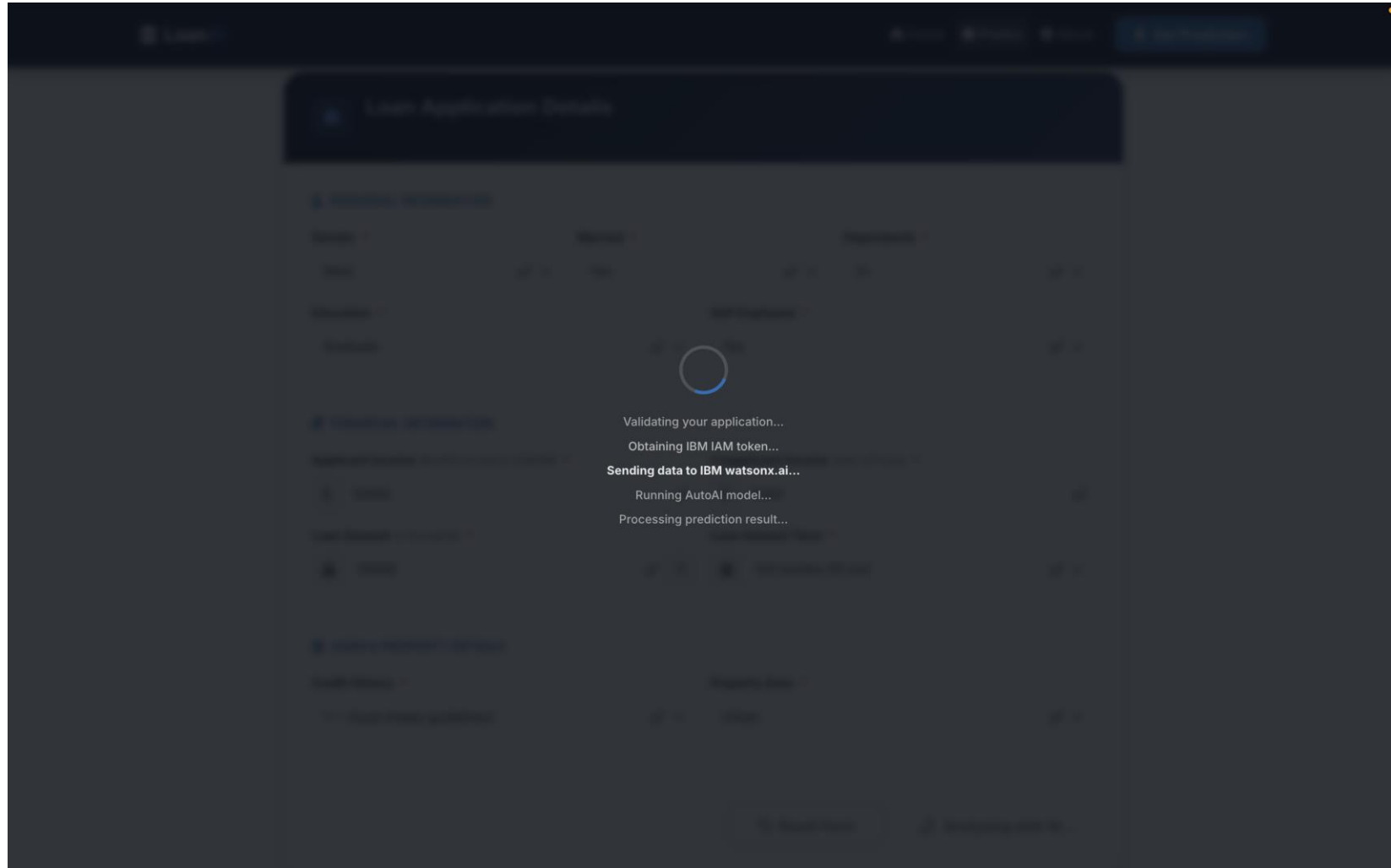
Property Area *
Urban

Reset Form

Predict Loan Approval

Training Screenshots

Website Working – Loan Application Form with Test Data 1 Progress



Training Screenshots

Website Working – Test Data 1 Prediction Result



HomePredictAbout

Get Prediction

Home / Predict / Result

 Prediction Result



Loan Approved

Congratulations! Based on your profile, the loan is likely to be approved.

AI Confidence Score66.73%

Loan ID: LP20260629212959Model: IBM watsonx.ai AutoAI

Submitted Application Details

PERSONAL		FINANCIAL	
Gender	Male	\$ Applicant Income	10000
Married	Yes	\$ Coapplicant Income	5000
Dependents	3+	Loan Amount	50000K
Education	Graduate	Loan Term	120 months
Self Employed	Yes	Credit History	Good (1)
		Property Area	Urban

Predict Again

Back to Home

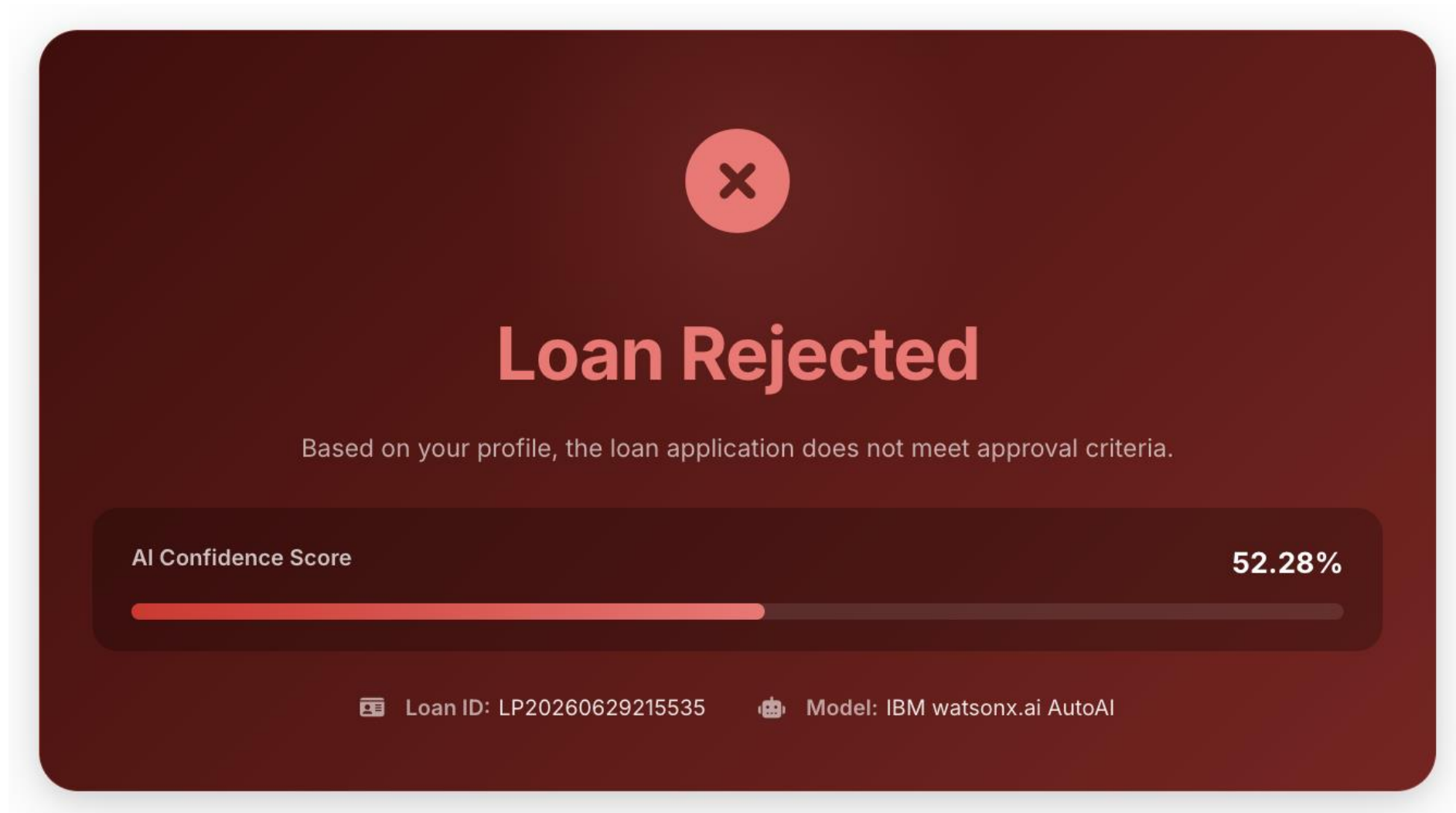
Training Screenshots

Website Working – Test Data Set 2

Submitted Application Details			
PERSONAL		FINANCIAL	
Gender	Male	\$ Applicant Income	1000
Married	Yes	\$ Coapplicant Income	0
Dependents	3+	Loan Amount	10000000000K
Education	Not Graduate	Loan Term	120 months
Self Employed	Yes	Credit History	Poor (0)
		Property Area	Semiurban

Training Screenshots

Website Working – Data Set 2 Prediction Result



Training Screenshots

Website Working – Website Details

 LoanAI

[Home](#) [Predict](#) [About](#) [Get Prediction](#)

 **Project Overview**

Loan Approval Prediction System is an end-to-end machine learning web application that uses an IBM watsonx.ai AutoAI model to predict whether a loan application will be **Approved** or **Rejected** based on 11 applicant features.

The model was trained on a real-world loan dataset and deployed as an Online Deployment on IBM Cloud (eu-de region). A Flask backend handles IAM authentication, input validation, and REST API communication — ensuring your credentials are never exposed to the browser.

Python 3.12

Flask 3.0

IBM Cloud

IBM AutoAI

Bootstrap 5

 **Dataset Info**

Source:

Loan Approval Prediction CSV

Target:

Loan_Status (Y / N)

Type:

Binary Classification

Features:

11 input features

Platform:

IBM watsonx.ai AutoAI

Region:

eu-de (Frankfurt)

Deployment:

Online (REST API)

 **Model Input Features**

 **Loan_ID**
Auto-generated unique ID (LP + timestamp). Not entered by user.

 **Gender**
Male / Female

 **Married**
Yes / No

 **Dependents**
0 / 1 / 2 / 3+

 **Education**
Graduate / Not Graduate

 **Self_Employed**
Yes / No

 **ApplicantIncome**
Monthly income of the primary applicant

 **CoapplicantIncome**
Monthly income of the co-applicant (0 if none)

 **LoanAmount**
Requested loan amount in thousands

Training Screenshots

Website Working – API Integration Flow

 LoanAI

[Home](#) [Predict](#) [About](#) [Get Prediction](#)

 **IBM Cloud & watsonx.ai**

IBM Cloud is a full-stack cloud platform with over 170 products and services. IBM watsonx.ai is the AI studio within IBM Cloud that provides tools to train, validate, tune, and deploy AI models with full governance.

The deployment endpoint used by this app is hosted in the **eu-de (Frankfurt)** region:

`eu-de.ml.cloud.ibm.com/ml/v4/deployments/.../predictions`

Authentication uses IBM IAM (Identity and Access Management) — the Flask backend exchanges your API key for a short-lived Bearer token automatically.

 **AutoAI Workflow**

- Data Ingestion** — Upload the CSV dataset to IBM watsonx.ai project.
- AutoAI Experiment** — Configure the target column (Loan_Status) and run.
- Pipeline Comparison** — AutoAI tries multiple algorithms (Random Forest, XGBoost, etc.) and hyper-parameter combinations.
- Best Pipeline** — The highest-ranked pipeline is saved as a model.
- Online Deployment** — Model is deployed as a REST endpoint.
- API Call** — This Flask app calls the endpoint with a JSON payload and receives `prediction + probability`.

 **API Integration Flow**

 **Step 1 — IAM Token**

Flask POSTs the `IBM_API_KEY` to `iam.cloud.ibm.com/identity/token` to obtain a Bearer token. Token is cached for ~1 hour.

 **Step 2 — Payload Build**

Form data is validated, numeric fields are cast, and Loan_ID is auto-generated. A structured JSON payload is assembled matching the model's expected schema.

 **Step 3 — Scoring Request**

Flask POSTs the payload to the IBM watsonx.ai deployment endpoint with the Bearer token in the Authorization header.

 **Step 4 — Parse & Display**

The response `prediction` (Y/N) and `probability` array are parsed. The result is stored in the Flask session and rendered on the result page.

Training Screenshots

Website Working – Project Structure



Project Structure

```
loan-approval-webapp/
├── app.py           ── Flask entry point, all routes
├── config.py        ── Environment configuration loader
├── requirements.txt  ── Python dependencies
├── .env.example      ── Template for environment variables
├── README.md        ── Full project documentation
├──
├── services/
│   ├── auth.py      ── IBM IAM token management
│   ├── predictor.py  ── IBM API payload + response parsing
│   └── utils.py      ── Validation, sanitization, Loan_ID gen
├──
├── templates/
│   ├── base.html    ── Shared navbar, footer, Bootstrap
│   ├── index.html   ── Home / landing page
│   ├── predict.html ── Loan prediction form
│   ├── result.html  ── Prediction result display
│   ├── about.html   ── Project & tech explanation
│   └── error.html    ── Friendly error page
├──
└── static/
    ├── css/style.css ── Custom banking theme styles
    └── js/main.js     ── Form validation & animations
```

[Try the Live Prediction](#)

Training Screenshots

Website Working – Test Data Set 3

PERSONAL

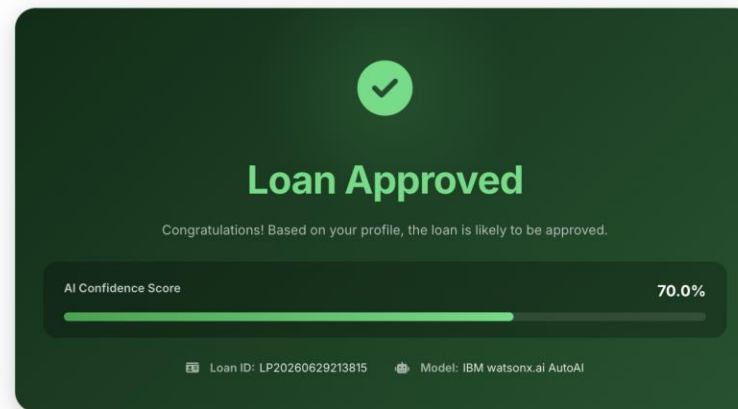
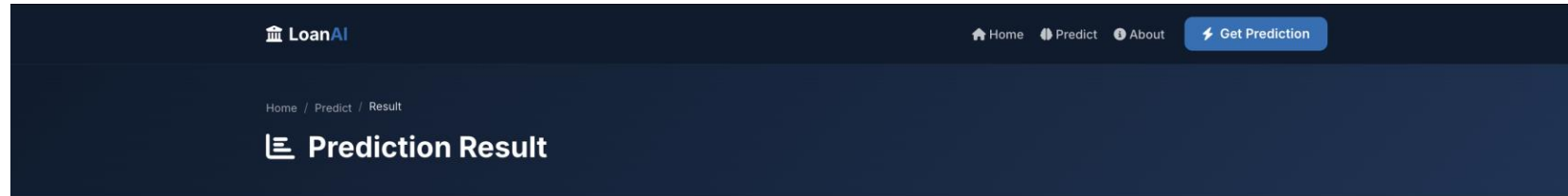
 Gender	Male
 Married	Yes
 Dependents	0
 Education	Not Graduate
 Self Employed	Yes

FINANCIAL

 Applicant Income	10000
 Coapplicant Income	0
 Loan Amount	30000K
 Loan Term	120 months
 Credit History	Good (1)
 Property Area	Rural

Training Screenshots

Website Working – Test Data Set 3 Prediction Result



Submitted Application Details			
PERSONAL		FINANCIAL	
Gender	Male	\$ Applicant Income	10000
Married	Yes	\$ Coapplicant Income	0
Dependents	0	Loan Amount	30000K
Education	Not Graduate	Loan Term	120 months
Self Employed	Yes	★ Credit History	Good (1)
		Property Area	Rural

Novelty and Uniqueness

- The solution automates the **loan approval prediction process**, reducing manual evaluation effort and improving decision-making efficiency.
- It uses **IBM watsonx.ai AutoAI** to automatically perform **data preprocessing, feature engineering, algorithm selection, and hyperparameter optimization** with minimal coding.
- The system evaluates **multiple machine learning pipelines** and automatically selects the **best-performing model** based on cross-validation accuracy.
- Automated feature engineering identifies hidden patterns and relationships among applicant attributes, improving prediction performance.
- The model provides **fast, consistent, and data-driven loan approval predictions**, helping financial institutions reduce human bias and improve operational efficiency.

Git Hub Link

- Git Hub Repo: <https://github.com/garigerahulkarunya-git/loan-approval-webapp.git>

Future Scope

- The solution can be integrated with **banking systems** to automate the end-to-end loan application screening and approval process.
- Additional applicant information, such as **credit scores, banking transactions, employment history, and financial records**, can be incorporated to improve prediction accuracy.
- Real-time integration with **credit bureaus and financial databases** can enhance applicant verification and fraud detection.
- Advanced AI models in **IBM watsonx.ai** can be used to continuously retrain and improve prediction performance as new loan data becomes available.
- The solution can be extended to **predict loan default risk, recommend suitable loan products, and provide explainable AI insights** to support transparent and data-driven lending decisions.

Certificates Earned: Getting Started with Artificial Intelligence

In recognition of the commitment to achieve
professional excellence



Rahul Karunya Garige

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Jun 23, 2026
Issued by: IBM SkillsBuild

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Certificates Earned: Lab Troubleshoot Your Code Using IBM Bob

IBM **SkillsBuild**

Completion Certificate



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Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 23 Jun 2026 (GMT)

Learning hours: 30 mins

Certificates Earned: Exploring Quantum Computing

IBM **SkillsBuild**

Completion Certificate



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Learning hours: 50 mins

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Has successfully satisfied the requirements for:

Journey to Cloud: Envisioning Your Solution



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Certificates Earned: Level Up Cybersecurity with Generative AI

IBM SkillsBuild

Completion Certificate



This certificate is presented to

Rahul Karunya Garige

for the completion of

Level Up Cybersecurity with Generative AI

(ALM-COURSE_3947352)

According to the Adobe Learning Manager system of record

Completion date: 26 Jun 2026 (GMT)

Learning hours: 1 hr 30 mins



Thank You!

Thank you for your time and interest.